Supplementary Material for "Exact Structural Abstraction and Tractability Limits"

Tristan Simas

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
FR1	`currentFamilies_partition_by_role` `Paper4dFrontier/Classification.lean`	FR2	`currentFamilies_partition_counts` `Paper4dFrontier/Classification.lean`
FR3	`current_families_have_finite_basis` `Paper4dFrontier/Classification.lean`	FR4	`current_positive_interfaces_factor_through_role` `Paper4dFrontier/Classification.lean`
FR5	`lifted_families_reduce_to_core` `Paper4dFrontier/Classification.lean`	FR6	`degenerateFamilies_complete` `Paper4dFrontier/Classification.lean`
FR7	`exists_decisionProblem_realizing_labeling` `Paper4dFrontier/Realizability.lean`	FR8	`realizingProblem_decisionEquiv_iff` `Paper4dFrontier/Realizability.lean`
FR13	`isOptimal_positiveAffineTransform_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR14	`opt_eq_positiveAffineTransform` `Paper4dFrontier/FamilyAxioms.lean`
FR15	`isSufficient_positiveAffineTransform_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR16	`isRelevant_positiveAffineTransform_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR17	`decisionEquiv_relabelActions_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR18	`decisionEquiv_relabelStates_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR19	`isSufficient_relabelActions_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR20	`isSufficient_relabelStates_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR21	`allProblems_relabelInvariant` `Paper4dFrontier/FamilyAxioms.lean`	FR22	`allProblems_kernelUniversal` `Paper4dFrontier/FamilyAxioms.lean`
FR23	`relabelInvariant_not_enough` `Paper4dFrontier/FamilyAxioms.lean`	FR24	`optRangeCard_realizingIdentity` `Paper4dFrontier/FamilyAxioms.lean`
FR25	`currentFamilies_explicit` `Paper4dFrontier/Classification.lean`	FR26	`coreFamilies_explicit` `Paper4dFrontier/Classification.lean`
FR27	`liftedFamilies_explicit` `Paper4dFrontier/Classification.lean`	FR28	`degenerateFamilies_explicit` `Paper4dFrontier/Classification.lean`
FR29	`primitiveMechanisms_explicit` `Paper4dFrontier/Classification.lean`	FR30	`isSufficient_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR31	`isRelevant_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR32	`isIrrelevant_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR33	`isMinimalSufficient_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR34	`sufficientSets_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR35	`relevantSet_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR37	`quotientCard_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR38	`quotientCard_realizingIdentity` `Paper4dFrontier/FamilyAxioms.lean`	FR39	`realizingProblemQuotientEquivLabelRange_apply_quotientMap` `Paper4dFrontier/Realizability.lean`

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR40	realizingProblem_quotientCard_eq_labelRangeCard Paper4dFrontier/Realizability.lean	FR41	certificationStatistic_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR42	sufficientSetCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR43	minimalSufficientSetCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR44	relevantCoordCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR45	setoidRealizingProblem_decisionEquiv_iff Paper4dFrontier/Realizability.lean
FR46	setoidRealizingProblemQuotientEquivSetoidQuotient_apply_quotientMap Paper4dFrontier/Realizability.lean	FR47	exists_decisionProblem_realizing_setoid Paper4dFrontier/Realizability.lean
FR48	isSufficient_iff_agreementRel_le_decisionEquiv Paper4dFrontier/FamilyAxioms.lean	FR49	isIrrelevant_iff_sufficient_erase Paper4dFrontier/FamilyAxioms.lean
FR50	isRelevant_iff_not_sufficient_erase Paper4dFrontier/FamilyAxioms.lean	FR51	quotientEquiv_of_decisionEquiv_iff_apply_quotientMap Paper4dFrontier/FamilyAxioms.lean
FR52	empty_sufficient_of_constant_opt Paper4dFrontier/ClosureLaws.lean	FR53	irrelevant_of_constant_opt Paper4dFrontier/ClosureLaws.lean
FR54	decisionEquiv_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR55	isSufficient_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean
FR56	isRelevant_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR57	decisionEquiv_duplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR58	isSufficient_duplicateState_iff Paper4dFrontier/ClosureLaws.lean	FR59	isRelevant_duplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR60	duplicateStateQuotientEquivOriginal_apply_quotientMap Paper4dFrontier/ClosureLaws.lean	FR66	booleanCube_card Paper4dFrontier/EnumerationBounds.lean
FR67	exists_booleanCube_larger_than_monomial Paper4dFrontier/EnumerationBounds.lean	FR68	decisionEquiv_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR69	isSufficient_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean	FR70	isRelevant_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR71	addDuplicateStateQuotientEquivOriginal_apply_quotientMap Paper4dFrontier/ClosureLaws.lean	FR72	some_mem_opt_addDuplicateAction_iff Paper4dFrontier/ClosureLaws.lean
FR73	none_mem_opt_addDuplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR83	isSufficient_noiseExtended_iff Paper4dFrontier/DimensionalNoiseExtension.lean
FR84	isRelevant_noiseExtended_iff Paper4dFrontier/DimensionalNoiseExtension.lean	FR85	lastCoord_irrelevant_noiseExtended Paper4dFrontier/DimensionalNoiseExtension.lean
FR100	parentTree_realTreewidth_one Paper4dFrontier/ParentTreewidth.lean	FR101	low_rank_only_witness Paper4dFrontier/LowRankOnlyWitness.lean
FR105	separable_only_witness Paper4dFrontier/SeparableOnlyWitness.lean	FR106	bounded_actions_only_witness Paper4dFrontier/BoundedActionsOnlyWitness2.lean
FR108	all_independent_core_mechanisms_nondegenerate Paper4dFrontier/Block2Progress.lean	FR110	parentTreeStructured_implies_treeStructured Paper4dFrontier/ParentTreewidth.lean
FR111	treeStructured_and_unique_parent_iff_parentTreeStructured Paper4dFrontier/ParentTreewidth.lean	FR112	bounded_treewidth_only_witness Paper4dFrontier/BoundedTreewidthOnlyWitness.lean
FR113	symmetry_only_witness Paper4dFrontier/SymmetryOnlyWitness.lean	FR114	weak_tree_structured_not_width_one Paper4dFrontier/LegacyTreeGap.lean
FR115	tree_structure_hardness_bridge Paper4dFrontier/TreeStructureHardnessBridge.lean	FR118	pairCrossDifference_eq_binaryCrossDifference_of_lt Paper4dFrontier/BinaryPairwiseDichotomy.lean
FR119	pairwise_zero_crossDifference_unaryDecomposition Paper4dFrontier/BinaryPairwiseDichotomy.lean	FR120	binary_pairwise_symmetry_dichotomy Paper4dFrontier/BinaryPairwiseDichotomy.lean
FR121	actionGapCrossDifference_eq_binaryCrossDifference_of_lt Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR122	decisionRelevant_zero_actionGap_implies_unaryReduction Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR123	binary_pairwise_symmetry_decision_relevant_dichotomy Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR124	block6_genuine_interaction_dichotomy_obstruction Paper4dFrontier/Block6Obstruction.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR125	decisionRelevantInteractionGraph_ addActionOffset Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR126	action_offset_can_force_constant_optimizer Paper4dFrontier/Block6Obstruction.lean
FR127	block6_action_offset_obstruction Paper4dFrontier/Block6Obstruction.lean	FR128	offsetNormalizedDecisionRelevantInteractionGraph_ wellDefined Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR129	block6_offset_normalized_obstruction Paper4dFrontier/Block6Obstruction.lean	FR130	neverOptimalGhost_decisionRelevantGraph_eq_top Paper4dFrontier/Block6Obstruction.lean
FR131	neverOptimalGhost_supportedGraph_eq_bot Paper4dFrontier/Block6Obstruction.lean	FR132	support_filtering_removes_known_block6_ obstructions Paper4dFrontier/Block6Obstruction.lean
FR133	block6_ghost_action_obstruction Paper4dFrontier/Block6Obstruction.lean	FR134	marginMasking_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean
FR135	marginMasking_zero_sufficient Paper4dFrontier/Block6Obstruction.lean	FR136	block6_optimizer_supported_obstruction Paper4dFrontier/Block6Obstruction.lean
FR137	MarginBounded Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR138	dominantPair_marginBounded Paper4dFrontier/Block6Obstruction.lean
FR139	dominantPair_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean	FR140	block6_margin_bounded_obstruction Paper4dFrontier/Block6Obstruction.lean
FR145	not_collapseLandscapeInfinity_of_oracle_ predicate Paper4dFrontier/MetaCharacterization.lean	FR176	DecisionQuotient.DecisionProblem.quotient_is_ coarsest paper4/DecisionQuotient/Quotient.lean
FR177	DecisionQuotient.DecisionProblem.quotient_has_ unique_factorization paper4/DecisionQuotient/Quotient.lean	FR178	DecisionQuotient.DecisionProblem.srank_eq_ relevant_card paper4/DecisionQuotient/Tractability/ StructuralRank.lean
FR179	DecisionQuotient.quotientEntropy_le_srank_ binary paper4/DecisionQuotient/Information.lean	FR180	DecisionQuotient.Statistics.fisherMatrix_rank_ eq_srank paper4/DecisionQuotient/Statistics/ FisherInformation.lean
FR181	DecisionQuotient.Information.compression_ below_srank_fails paper4/DecisionQuotient/Information/ RDSrank.lean	FR182	DecisionQuotient.Information.srank_bits_ sufficient paper4/DecisionQuotient/Information/ RDSrank.lean
FR183	DecisionQuotient.Physics.single_future_zero_ cost paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	FR184	DecisionQuotient.Physics.transportCost_pos_of_ offDiag paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean
FR185	no_closureInvariant_predicate_of_orbit_gap Paper4dFrontier/ObstructionPredicateCandidates.lean	FR186	no_admissibleNormalizationPredicate_decides_ dominantPair Paper4dFrontier/ObstructionPredicateCandidates.lean
FR187	no_admissibleNormalizationPredicate_decides_ marginBounded Paper4dFrontier/ObstructionPredicateCandidates.lean	FR188	no_admissibleNormalizationPredicate_decides_ ghostActionTwoPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean
FR189	no_admissibleNormalizationPredicate_decides_ offsetActionZeroPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean	FR190	admissibleCollapseLandscapeInfinity_full_paper Paper4dFrontier/ObstructionPredicateCandidates.lean
FR191	ClosureSoundPackage Paper4dFrontier/ObstructionPredicateCandidates.lean	FR192	ReasonableGuardrailPackage Paper4dFrontier/ObstructionPredicateCandidates.lean
FR193	no_closureSoundPackage_predicate_decides_four_ obstruction_families Paper4dFrontier/ObstructionPredicateCandidates.lean	FR194	no_reasonableGuardrailPackage_predicate_ decides_four_obstruction_families Paper4dFrontier/ObstructionPredicateCandidates.lean
FR197	classifier_agrees_on_closureEquivalent_of_ correctOnDomain Paper4dFrontier/AdmissibleCharacterization.lean	FR198	no_correctOnDomain_classifier_of_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean
FR199	correct_classifier_inherits_ closureLawInvariant Paper4dFrontier/AdmissibleCharacterization.lean	FR200	admissibleNormalizationPredicate_has_explicit_ inhabitants Paper4dFrontier/AdmissibleCharacterization.lean
FR201	closureLawInvariant_of_iff_of_ closureEquivalent Paper4dFrontier/AdmissibleCharacterization.lean	FR202	exists_orbit_gap_of_not_closureLawInvariant Paper4dFrontier/AdmissibleCharacterization.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR203	closureLawInvariant_iff_no_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean	FR204	no_exact_closureLawInvariant_classifier_iff_ exists_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean
FR205	distinctActionCount Paper4dFrontier/DistinctActionProfiles.lean	FR206	actionCount_profileCompressedSlice Paper4dFrontier/DistinctActionProfiles.lean
FR207	decisionEquiv_profileCompressedSlice_iff Paper4dFrontier/DistinctActionProfiles.lean	FR208	profileCompressedSlice_preserves_ exactCertification Paper4dFrontier/DistinctActionProfiles.lean
FR209	profileCompressedSlice_bounded_actions Paper4dFrontier/DistinctActionProfiles.lean	FR210	OrbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean
FR211	no_orbitGapOn_of_exact_classifiable_by_ closureLawInvariant_onDomain Paper4dFrontier/AdmissibleCharacterization.lean	FR212	exact_classifiable_by_closureLawInvariant_ onDomain_iff_no_orbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean
FR213	no_exact_closureLawInvariant_classifier_ onDomain_iff_orbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean	FR214	boundedPatternScheme_holds_largeActionCount_ iff Paper4dFrontier/AdmissibleCharacterization.lean
FR215	boundedPatternDefinable_eventually_constant_ in_actionCount Paper4dFrontier/AdmissibleCharacterization.lean	FR216	boundedPatternDefinable_largeActionCount_ agrees Paper4dFrontier/AdmissibleCharacterization.lean
FR217	payloadSufficient_iff_realizingProblem_ isSufficient Paper4dFrontier/Realizability.lean	FR218	payloadRelevant_iff_realizingProblem_ isRelevant Paper4dFrontier/Realizability.lean
FR219	payloadIrrelevant_iff_realizingProblem_ isIrrelevant Paper4dFrontier/Realizability.lean	FR220	payloadMinimalSufficient_iff_realizingProblem_ isMinimalSufficient Paper4dFrontier/Realizability.lean
FR221	payloadSufficientSets_eq_realizingProblem_ sufficientSets Paper4dFrontier/Realizability.lean	FR222	payloadRelevantSet_eq_realizingProblem_ relevantSet Paper4dFrontier/Realizability.lean
FR223	feasiblePayloadSufficient_iff_ lawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR224	feasiblePayloadRelevant_iff_ lawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR225	feasiblePayloadIrrelevant_iff_ lawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR226	setValuedPayloadSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR227	setValuedPayloadRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR228	setValuedPayloadIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR229	outputSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR230	outputSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR231	outputSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR232	approximationSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR233	approximationSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR234	approximationSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR235	outputSemanticsSufficientSets_eq_of_ outputSemanticsEquivalent Paper4dFrontier/Realizability.lean	FR236	outputSemanticsRelevantSet_eq_of_ outputSemanticsEquivalent Paper4dFrontier/Realizability.lean
FR237	quotientMap_realizes_setoid_asOutputSemantics Paper4dFrontier/Realizability.lean	FR240	exists_outputSemantics_realizing_setoid Paper4dFrontier/Realizability.lean
FR241	outputSemanticsSetoid Paper4dFrontier/Realizability.lean	FR242	SemanticallyExtensionalMap Paper4dFrontier/Realizability.lean
FR243	semanticallyExtensionalMap_factors_through_ outputSemanticsQuotient Paper4dFrontier/Realizability.lean	FR244	semanticallyExtensionalClaim_factors_through_ outputSemanticsQuotient Paper4dFrontier/Realizability.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	**Lean handle**
Corollary 6.25: Exact Approximation Semantics Transfer	FR234, FR233, FR232
Corollary 6.6: Constant-Optimizer Collapse Forces Trivial Certification	FR52, FR53
Corollary 6.27: Every State Equivalence Relation Is Realizable as Exact Output Semantics	FR240, FR237
Corollary 2.3: Finite Basis for the Current Positive Landscape	FR3, FR29
Corollary 4.11: No Correct Tractability Classifier	FR197, FR199, FR193, FR198
Corollary 4.13: Orbit-Gap Completeness on Closure-Closed Domains	FR210, FR212, FR213, FR211
Corollary 5.4: Realizability Alone Cannot Be the Frontier	FR7, FR8
Corollary 4.10: Reasonable Guardrail Packages	FR194
Corollary 6.24: Arbitrary Exact Output Semantics Transfer	FR231, FR230, FR229
Corollary 6.3: Relevance Is Failure of Erased Sufficiency	FR49, FR50
Corollary 6.22: Nonempty Set-Valued Payload Transfer	FR225, FR224, FR223
Corollary 6.4: Certification Statistics Factor Through the Quotient Relation	FR41, FR43, FR44, FR42
Corollary 6.2: Sufficiency Is Relation Refinement	FR48
Corollary 6.23: Arbitrary Set-Valued Payload Transfer via Failure Tokens	FR228, FR227, FR226
Proposition 6.18: The Admissibility Layer Is Nonempty	FR200
Proposition 6.8: Positive Affine Utility Reparameterizations Are Invisible	FR13, FR16, FR15, FR14
Proposition 4.1: Binary Pairwise Symmetry Dichotomy	FR120, FR118, FR119
Proposition 6.20: Bounded-Pattern Predicates Stabilize Above a Finite Action Bound	FR215, FR216, FR214
Proposition 6.15: Closure Operations Preserve Exact Certification	FR57, FR17, FR18, FR60, FR56, FR59, FR84, FR16, FR55, FR58, FR83, FR15, FR19, FR20, FR85
Proposition 4.2: Decision-Relevant Binary Pairwise Dichotomy	FR121, FR123, FR124, FR122
Proposition 3.2: Degenerate Positive Cases	FR1, FR6
Proposition 6.1: Exact Certification Depends Only on the Decision Quotient Relation	FR32, FR33, FR31, FR30, FR37, FR51, FR35, FR34
Proposition 6.19: Bounded Distinct Action Profiles Compress to Bounded Actions	FR206, FR207, FR205, FR209, FR208
Proposition 6.9: Duplicate Actions and Duplicate States Preserve Certification	FR71, FR68, FR54, FR70, FR56, FR69, FR55, FR73, FR72
Proposition 6.7: Explicit Enumeration Is Parameter-Dependent, Not Structural	FR66, FR67
Proposition 2.4: Explicit Primitive Basis	FR29
Proposition 2.2: Explicit Inventory	FR26, FR25, FR3, FR28, FR27
Proposition 2.5: All Independent Core Mechanisms Are Nondegenerate	FR108, FR106, FR112, FR101, FR105, FR113
Proposition 2.9: Current Positive Interfaces Factor Through Role Classification	FR4
Proposition 6.12: Invariance Alone Is Too Weak	FR22, FR21, FR23
Proposition 2.6: Weak Tree Ordering Does Not Imply Width One	FR115, FR114
Proposition 3.1: Lifted Cases Introduce No New Primitive	FR5
Proposition 6.14: Unrestricted Predicates Trivialize Exact Characterization	FR145

Paper claim	**Lean handle**
Proposition 4.12: Orbit-Gap Completeness for Exact Classification	`FR203`, `FR201`, `FR202`, `FR204`
Proposition 4.3: Orbit-Gap Template	`FR185`
Proposition 2.8: Parent-Tree Structure Gives Width-One Decompositions	`FR110`, `FR100`, `FR111`
Proposition 6.21: Deterministic Payload Transfer	`FR219`, `FR220`, `FR222`, `FR218`, `FR221`, `FR217`
Proposition 6.11: Quotient Size Is Unbounded Under Realizability	`FR24`, `FR38`
Proposition 5.3: The Realized Quotient Is Exactly the Label Range	`FR39`, `FR40`
Proposition 6.10: Relabeling Invariance Is Forced by Exact Certification	`FR17`, `FR18`, `FR19`, `FR20`
Proposition 6.26: Exact Semantic Claims Depend Only on Admissible-Output Equivalence	`FR236`, `FR235`
Proposition 6.28: Semantically Extensional Claims Factor Through the Exact-Semantics Quotient	`FR242`, `FR241`, `FR244`, `FR243`
Proposition 5.2: Every Equivalence Relation Is Optimizer-Realizable	`FR47`, `FR46`, `FR45`
Proposition 6.13: Independent Summary Frameworks Converge on the Same Structural Core	`FR178`, `FR181`, `FR182`, `FR180`, `FR179`
Proposition 2.7: Cyclic Dependencies Recover Hardness	`FR110`, `FR100`, `FR111`, `FR115`
Proposition 6.5: Zero-Distortion Summaries Refine the Optimizer Quotient	`FR177`, `FR176`
Theorem 4.8: Admissible Collapse Across the Four Obstruction Families	`FR190`
Theorem 4.9: Closure-Sound Package No-Go	`FR193`
Definition 6.17: Admissible Normalization Predicate	`FR197`, `FR199`, `FR198`
Theorem 2.1: Explicit Partition of the Current Positive Landscape	`FR26`, `FR25`, `FR1`, `FR2`, `FR28`, `FR27`
Theorem 4.4: Dominant-Pair Admissible No-Go	`FR186`
Theorem 4.6: Ghost-Action Admissible No-Go	`FR188`
Theorem 5.1: Every Labeling Kernel Is Optimizer-Realizable	`FR7`, `FR8`
Theorem 4.5: Margin-Masking Admissible No-Go	`FR187`
Theorem 4.7: Offset Admissible No-Go	`FR189`

Auto summary: mapped 55/55 (full=55, derived=0, unmapped=0).